

# 240 Gbit/s Coherent PON Demo with Simplified ONU Design Using Uncalibrated Lasers and Coexistence with Legacy PONs

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**Abstract** We demonstrate in-service alignment of a wavelength-uncalibrated LO laser in a proof-of-concept coherent PON setup, enabling a 240 Gbit/s downstream link. Coexistence with legacy PON validates a potential migration path toward future high-capacity access networks. ©2026 The Author(s)

## Introduction

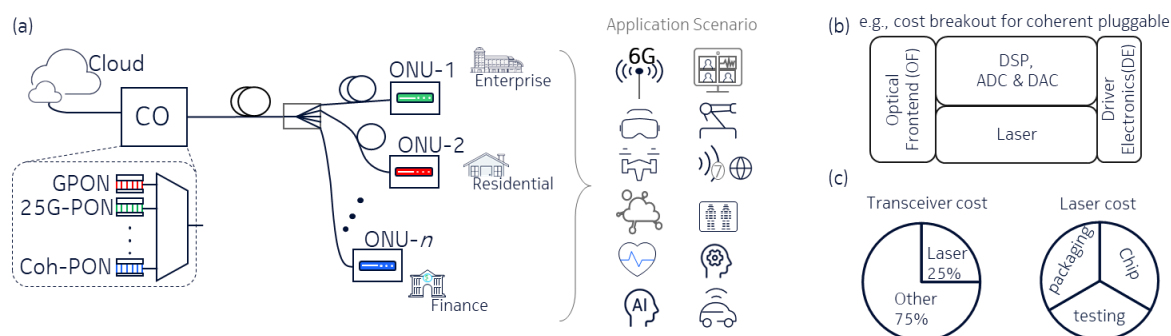
The evolution of passive optical networks (PON) is driven by increasing operator demands and guided by standardization bodies. Currently, ITU-T Study Group 15, Question 2, is exploring various technology options under the framework of “Very High Speed PON (VHSP)” [1]. Among these candidates, complex-valued higher-order modulation combined with coherent detection is considered a possible solution [1-2]. In addition, C-band operation of coherent PON systems enables a more relaxed O-band wavelength plan and simplifies coexistence filter design. Figure 1(a) illustrates the envisioned coexistence scenario and application landscape, ranging from traditional broadband services to emerging artificial intelligence (AI) enabled applications.

Figure 1(b) illustrates the cost breakdown of coherent transceiver hardware, including the optical frontend, digital signal processor (DSP), driver electronics, and a fully calibrated, wavelength-stable, narrow-linewidth laser source [3]. While such architectures are well established in metro and long-haul systems, their adoption in data center applications has only recently begun. However, the direct application of coherent transceivers in PON remains challenging. Unlike point-to-point systems, PON operates in a point-to-multipoint configuration with inherent cost asymmetry, where the optical line terminal (OLT) cost is shared among multiple users, making low-cost

optical network units (ONUs) essential [4]. Consequently, coherent PON solutions must be carefully optimized and simplified to reduce cost, packaging, testing, and power consumption across all major transceiver components, particularly at the ONU [5-6].

As shown in Fig. 1(c), the laser contributes approximately 25% of the total transceiver cost [7]. Although simpler laser technologies, such as distributed feedback (DFB) lasers, can reduce chip and packaging costs, which account for approximately two-thirds of the overall laser cost, see Fig. 1(c), bottom-right pie chart, testing costs remain substantial and do not scale effectively with production volume. Therefore, wavelength-uncalibrated ( $\lambda$ -UC) lasers are desirable for PONs. However, these lasers exhibit significant wavelength uncertainty, often several nanometers, leading to an inherent wavelength mismatch between the local oscillator (LO) and the received signal, which effectively prevents coherent detection. Consequently, new coherent ONU receiver architectures based on  $\lambda$ -UC lasers and in-service alignment techniques are required for PON adaptation. In [8], we demonstrated the blind locking of  $\lambda$ -UC LO lasers in a laboratory setup using off-the-shelf components.

In this demo, we present a 240 Gbit/s coherent PON downstream (DS) demonstrator using autonomous wavelength locking of  $\lambda$ -UC lasers during ONU startup. The system further



**Fig. 1:** (a) Future PON coexistence architecture and application scenarios. (b) Cost breakdown of a coherent transceiver, (c) highlighting the significant contribution of the laser.

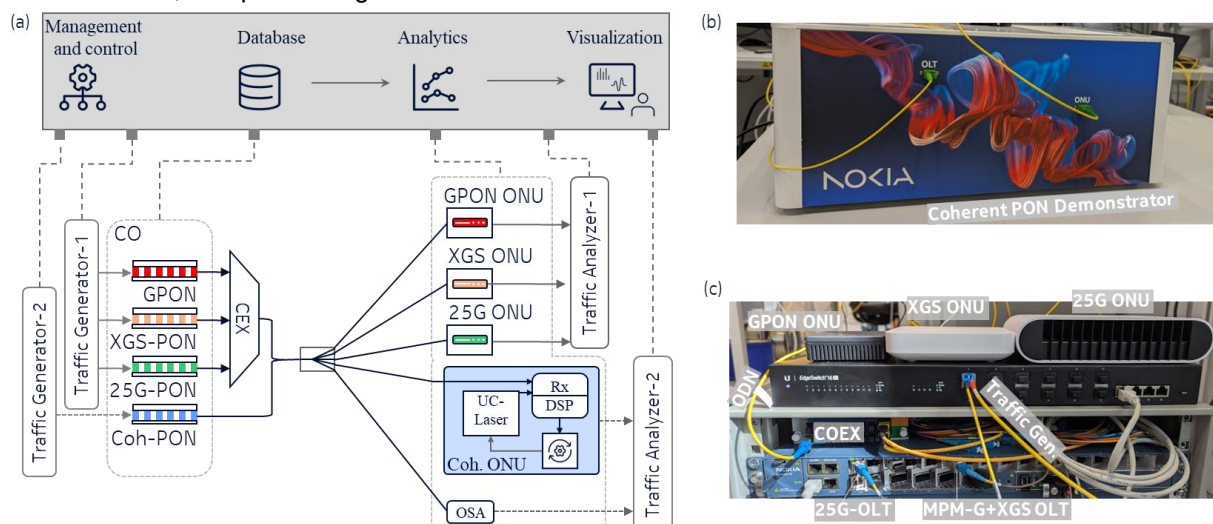
demonstrates coexistence with legacy PON technologies (GPON, XGS-PON, and 25G-PON) over a shared optical distribution network (ODN), highlighting a potential migration path toward future high-capacity access networks.

### Simplified ONU: Using Uncalibrated Laser

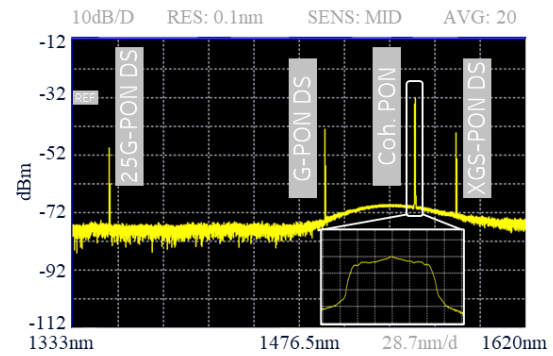
We propose a coherent ONU architecture utilizing  $\lambda$ -UC lasers used as the LO. In this approach, the LO laser initially operates at an arbitrary wavelength within its emission band. The ONU performs a temperature sweep of the LO laser while monitoring the received electrical signal power. The signal power reaches a maximum when the LO wavelength is coarsely aligned with the DS signal. Subsequently, DSP feedback is employed for fine frequency alignment, bringing the LO within the required operating range. This combined coarse and fine alignment procedure enables reliable wavelength acquisition without prior laser calibration and ensures stable operation during normal transmission.

### 240Gbit/s Coherent PON Demonstration and Coexistence

Figure 2(a) illustrates the system-level architecture of the 240 Gbit/s coherent PON demonstrator, including coexistence with legacy PON generations. At the central office (CO), a commercial traffic generator (Traffic Generator 1) provides service traffic for GPON, XGS-PON, and 25G-PON OLTs, while a custom-built traffic generator (Traffic Generator 2) produces downstream test sequences for the coherent PON OLT. Legacy PON systems employ commercial OLT pluggables integrated with a cloud-based network management platform. For the coherent PON, an in-house demonstrator is used, with the OLT and ONU implemented within the same housing (see Fig. 2(b)), and MATLAB-based software enabling data extraction, and processing.



**Fig. 2:** 240 Gbit/s coherent PON demonstration with multi-PON coexistence. (a) System-level architecture including CO, shared ODN, legacy ONUs, and a coherent ONU with UC laser and real-time analytics framework. (b) Coherent PON demonstrator. (c) Experimental coexistence setup with GPON, XGS-PON, and 25G-PON systems.



**Fig. 3:** Optical spectrum in the coexistence scenario.

All downstream signals are combined at the CO via a coexistence element (CEX) and a 2×1 fiber coupler before transmission over a shared ODN. At the receiver side, commercial off-the-shelf ONUs (GPON, XGS-PON, and 25G-PON) operate in parallel with the coherent ONU demonstrator, as shown in Fig. 2(b-c). All ONUs are connected to the management and analytics framework via two traffic analyzers: a commercial platform (Traffic Analyzer 1) and a custom-built system (Traffic Analyzer 2).

The demonstrator is further integrated with higher-layer functionalities, including management and control, database, analytics, and visualization (top layer in Fig. 2(a)), allowing comprehensive monitoring and performance evaluation. Figures 2(b) and 2(c) show the physical realization of the demonstration platform, integrating traffic generation, coexistence modules, and GPON/XGS-PON/25G-PON transceivers interconnected through off-the-shelf optical components and the coherent PON demonstrator. Figure 3 presents the measured optical spectrum of the coexistence scenario, confirming interference-free operation among the deployed PON systems.

### Visualization and Performance Evaluation

For live performance evaluation, the end-to-end behavior of the coherent PON demonstrator is monitored by capturing emulated traffic and extracting key performance indicators. Figure 4 presents the visualization interface, which displays wavelength alignment status during ONU startup and steady-state operation, including high-level indicators such as *coarse alignment*, *fine alignment*, *locked*, and *tracking*.

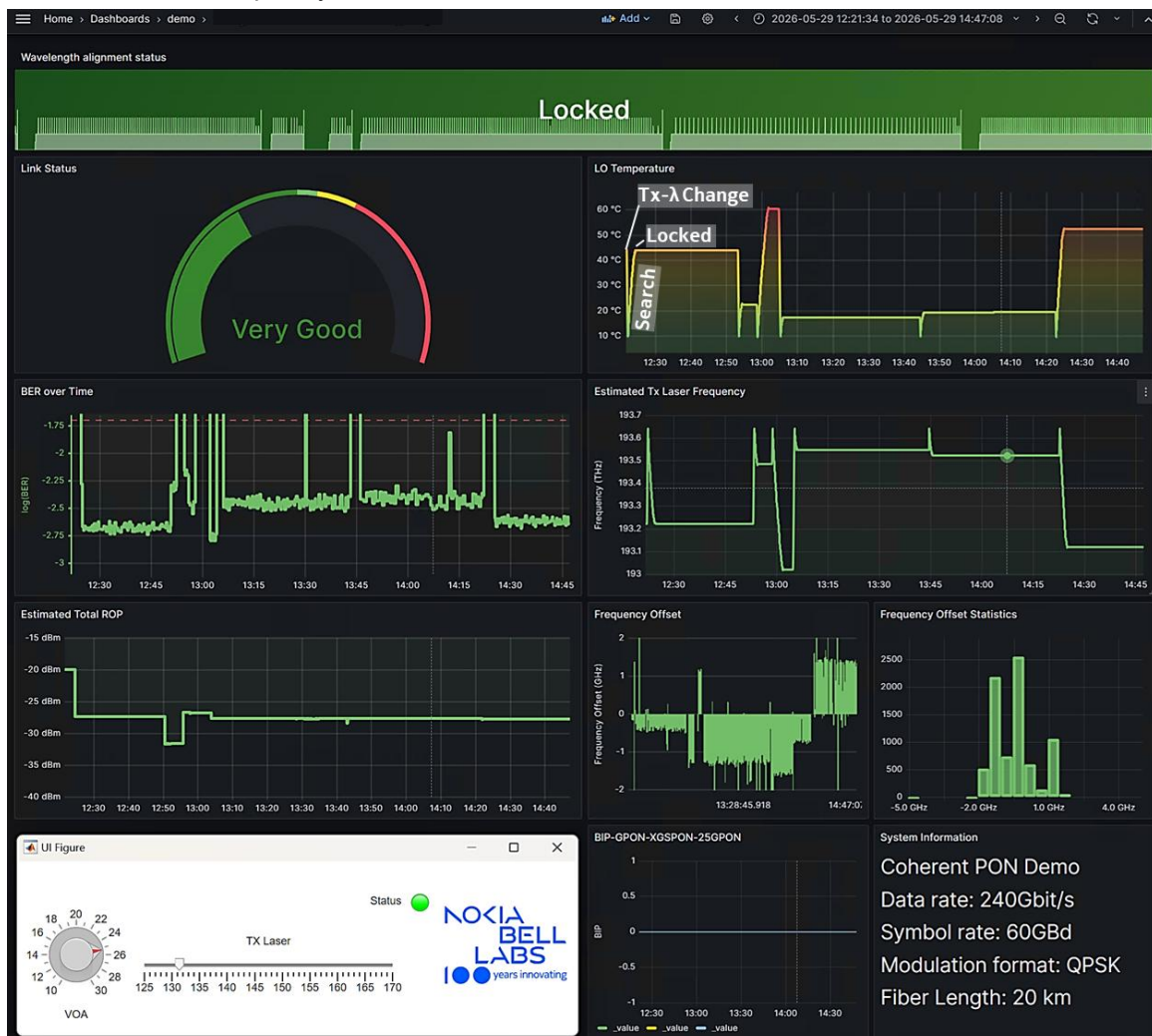
The LO-temperature panel illustrates the blind LO wavelength acquisition process over time. Initially, an intentional transmitter wavelength change ("Tx- $\lambda$  Change") is introduced, after which the LO enters the coarse alignment phase and sweeps across its tuning range (labeled "Search"). After successful alignment, the LO transitions into the locking and tracking phase (labeled "Locked"), indicating stable wavelength alignment. Further intentional Tx- $\lambda$  changes demonstrate autonomous re-acquisition behavior within the same trace. Additional metrics include bit-error ratio (BER) over time, estimated total received optical power (ROP), and link status. The estimated Tx laser frequency is tracked to reflect

wavelength variations during operation. The frequency-offset panel shows temporal evolution, while the corresponding histogram provides the statistical distribution of the residual carrier-frequency offset.

For legacy PON systems (GPON, XGS-PON, and 25G-PON), integrity indicators such as Bit Interleaved Parity (BIP) errors are plotted, although only minimal activity is observed due to the absence of interference from the coherent system. The platform further allows emulation of different ODN loss conditions and Tx laser wavelength variations, enabling observation of the corresponding LO locking dynamics.

### Conclusions

In this demonstration, we present a 240 Gbit/s coherent PON system using a wavelength-uncalibrated laser. The proposed autonomous LO locking enables calibration-free ONU operation. Co-existence with GPON, XGS-PON, and 25G-PON is validated, demonstrating the feasibility of integrating coherent PON into existing access networks.



**Fig 4:** Live visualization interface for coherent PON performance monitoring, showing wavelength alignment status, LO temperature evolution during acquisition, BER, received optical power, frequency offset, and link status.

## Acknowledgements

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